

Software Requirements Specification (SRS)

Project: Bank Management System (BMS)

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Date: 09-09-2026

Status: Draft for Review

Revision History

Version	Date	Author	Change Summary	Approval
1.0	09-09-2026	Aditi, Aarush, Sai Dhyan, Abhiroop	Initial SRS drafted from course template	Pending

Approvals

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Table of Contents

1. Introduction	3
2. Overall description	4
3. External interface requirements	5
4. System features (detailed)	6
5. Non-functional requirements (detailed)	10
6. Quality attributes & Acceptance tests	12
7. System models and diagrams.....	13
8. Requirements Traceability Matrix (RTM)	15

1. Introduction

1.1 Purpose

This document describes the requirements for the Bank Management System (BMS) being developed as part of our coursework. It covers what the system should do, the interfaces it will use, the security and performance requirements, and how each requirement will be tested. The main goal is to bring common retail banking activities that are usually handled at a branch counter or recorded manually into one web-based application. The system will support customers, bank staff, and administrators. This SRS will be used by the development and testing teams, as well as our project guide and evaluators, throughout the project.

1.2 Scope

The current version of the BMS supports customer registration and login, account opening with KYC verification, cash deposits and withdrawals, transfers between accounts within the same bank, loan applications and their approval process, mini-statements and receipt/passbook generation, teller-assisted counter operations, and administrative tasks such as user and role management, reports, and audit logs. The system uses its own database and does not connect to external interbank networks such as NEFT, RTGS, or SWIFT in this release. These integrations can be considered for a future version. Physical banking equipment such as ATMs, card readers, and cash dispensers is also outside the scope of this project.

1.3 Audience

This document is intended for the project team, including developers, testers, database administrators, and bank operations staff involved in reviewing the workflows. It is also meant for our project guide and evaluators, who will use it to understand and assess the system requirements.

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
BMS	Bank Management System
UI	User Interface
API	Application Programming Interface
OTP	One-Time Password
KYC	Know Your Customer
DB	Database
RBAC	Role-Based Access Control
TLS/SSL	Transport Layer Security / Secure Sockets Layer
PAN	Permanent Account Number (customer identity document)
RTM	Requirements Traceability Matrix

2. Overall description

2.1 Product perspective

The BMS is designed to reduce the need for manual, ledger-based branch operations by bringing them into one system. It has three main parts that use the same database: a customer portal, a teller/back-office application, and an administrator console. The system is designed as a standalone application, with separate modules for Authentication, Accounts, Transactions, Loans, and Reporting. Keeping these modules separate will also make it easier to add features such as a payment gateway or interbank connectivity in a future release.

2.2 Major product functions

- Customer registration, login, and multi-factor authentication
- Account opening with KYC document capture and verification
- Cash deposit and withdrawal recording against customer accounts
- Intra-bank funds transfer between own or other customers' accounts
- Loan application, document upload, and approval/rejection workflow
- Mini-statement, passbook, and transaction receipt generation
- Staff-assisted counter/teller operations for walk-in customers
- Administrative user, role, and branch management
- Report generation for management (daily transactions, loan status, account summaries)
- Audit logging and monitoring of all sensitive operations

2.3 User roles and characteristics

Role	Characteristics
Customer	General public with basic digital literacy; expects a simple, fast, and secure interface for everyday banking tasks.
Bank Teller	Branch staff who assist walk-in customers with deposits, withdrawals, and account opening; needs quick data-entry screens with validation.
Branch Manager	Approves loans and account exceptions above teller authority; needs summary dashboards and reports.
System Administrator	Manages users, roles, branches, and system configuration; needs full audit visibility and configuration control.

2.4 Operating environment

The BMS will be available as a web application and should work on modern browsers such as Chrome, Edge, and Firefox on both desktop and mobile devices. The backend will use an application server, such as Node.js, Java, or .NET, connected to a relational database such as MySQL, PostgreSQL, or SQL Server. The system can be hosted on-premises or in the cloud. Branch terminals will need a reasonably stable connection to the central server because teller operations depend on access to the shared system.

2.5 Constraints

- Must comply with applicable data protection and banking regulations for customer financial data.

- Password and PIN storage must use approved one-way hashing/encryption; card-like identifiers (e.g., PAN, account numbers) must never be stored or displayed in plaintext where avoidable.
- Must be developed using technologies approved by the course/project guide.
- Development timeline is constrained to a single academic semester.

3. External interface requirements

3.1 User interfaces

The system will use a responsive web interface, with separate menus and dashboards for Customers, Tellers, Managers, and Administrators. This avoids giving every user the same screen and helps each role reach the functions they actually need. The interface should also support accessibility features such as screen-reader-friendly labels and error messages, along with a high-contrast mode for users with visual impairments.

3.2 Hardware interfaces

- Standard workstation/PC or mobile device with keyboard, mouse/touch input, and display.
- Optional receipt/passbook printer at branch teller counters (vendor-specific driver interface).
- Optional barcode/QR scanner for KYC document intake at branch counters.

3.3 Software interfaces

- Central Database Service — stores customer, account, transaction, and loan data via secure DB connections.
- Email/SMS Gateway API — sends OTPs, transaction alerts, and statement notifications.
- Authentication/Identity Service — validates login credentials and issues session tokens.
- Optional future interface: external payment gateway or interbank settlement API (out of current scope, reserved for future release).

3.4 Communications

- TLS 1.2 or higher enforced for all client-server communication.
- Session tokens expire after a configurable idle timeout (default 10 minutes) and require re-authentication.
- All API requests and responses are exchanged as JSON over HTTPS.

4. System features (detailed)

Each requirement in this section includes acceptance criteria and a related test case so that the team has a clear way to check whether it has been implemented correctly. The requirements are numbered from BMS-F-001 onward so they can be easily referenced in the rest of the document.

4.1 User Authentication & Account Management

This module handles customer registration and login, along with session management and role-based access. It ensures that users can sign in securely and only access the features available to their role.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test ref	Comments/Dependencies
BMS-F-001	The system shall let a new customer register using their name, email/mobile number, and government ID number, while checking that these details are not already registered.	Functional	High	Customer	AC-001: Duplicate ID/email is rejected with a clear message. Test: TC-Auth-01	Requires KYC module
BMS-F-002	The system shall authenticate users using their username/customer ID and password, followed by an OTP sent to their registered mobile number or email.	Functional	High	Security	AC-002: Correct credentials + valid OTP grants an authenticated session. Test: TC-Auth-02	Requires SMS/Email gateway
BMS-F-003	The system shall lock a user account after 5 consecutive unsuccessful login attempts and record the event in the audit log.	Functional	High	Security	AC-003: 6th failed attempt blocks login and displays lockout message. Test: TC-Auth-03	Unlock requires admin or timed cooldown
BMS-F-004	The system shall use role-based access control (RBAC) so that Customers, Tellers, Managers, and Admins can access only the functions assigned to their roles.	Functional	High	Security	AC-004: Attempting an out-of-role action returns 'access denied'. Test: TC-Auth-04	Applies to every module

4.2 Account Opening & KYC

This module collects the customer's personal details and KYC documents and uses them to create a new account record after the required verification steps.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test ref	Comments/Dependencies
BMS-F-005	The system shall allow a Teller or customer to open a new savings or current account by entering the required personal details and uploading KYC documents.	Functional	High	Business	AC-005: New account record is created with a unique account number. Test: TC-KYC-01	Requires document upload
BMS-F-006	The system shall send newly opened accounts to a Branch Manager for KYC verification and approval before they are activated.	Functional	High	Compliance	AC-006: Account status remains 'Pending' until Manager approves. Test: TC-KYC-02	Depends on BMS-F-005
BMS-F-007	The system shall notify the customer by email or SMS when the account has been approved and activated.	Functional	Medium	Customer	AC-007: Notification is sent within 1 minute of approval. Test: TC-KYC-03	Requires notification gateway

4.3 Cash Deposit & Withdrawal

This module handles routine cash deposits and withdrawals. It checks the account status and available balance before completing a transaction and updates the balance once the transaction is successful.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test ref	Comments/Dependencies
BMS-F-008	The system shall allow a Teller to record a cash deposit for a valid, active account and update the account	Functional	High	Operations	AC-008: Balance increases by exact deposit amount; receipt is generated.	Ledger consistency required

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test ref	Comments/Dependencies
	balance immediately.				Test: TC-Txn-01	
BMS-F-009	The system shall check the available balance and daily withdrawal limit before approving a withdrawal.	Functional	High	Business	AC-009: Withdrawal is rejected if amount exceeds balance or daily limit. Test: TC-Txn-02	Limits configurable by Admin
BMS-F-010	The system shall generate a unique transaction receipt/reference number for every deposit and withdrawal.	Functional	Medium	Customer	AC-010: Every transaction has a unique, retrievable reference ID. Test: TC-Txn-03	Feeds into RTM/audit

4.4 Funds Transfer

This module allows money to be transferred between accounts within the bank. For higher-value transfers, the system uses OTP verification, and a transaction receipt is generated after the transfer is completed.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test ref	Comments/Dependencies
BMS-F-011	The system shall allow a customer to transfer funds between their own accounts or to another valid account within the bank.	Functional	High	Customer	AC-011: Transfer succeeds only if sender balance is sufficient. Test: TC-Xfer-01	Requires OTP confirmation
BMS-F-012	The system shall require OTP confirmation for transfers above a configurable threshold amount.	Functional	High	Security	AC-012: Transfers above threshold are blocked without valid OTP. Test: TC-Xfer-02	Depends on BMS-F-002
BMS-F-013	The system shall reject transfers that exceed the configured daily	Functional	Medium	Business	AC-013: Over-limit transfer attempt is rejected with an explanatory	Limits configurable by Admin

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test ref	Comments/Dependencies
	transfer limit and show the reason to the user.				message. Test: TC-Xfer-03	

4.5 Loan Management

This module allows customers to submit loan applications and supporting documents. Bank staff can then review the applications and either approve or reject them, along with recording the relevant remarks.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test ref	Comments/Dependencies
BMS-F-014	The system shall allow a customer to submit a loan application with the requested amount, tenure, purpose, and supporting documents.	Functional	Medium	Customer	AC-014: Application is saved with status 'Submitted' and a reference ID. Test: TC-Loan-01	Requires document upload
BMS-F-015	The system shall allow a Branch Manager to review a loan application and approve or reject it, with remarks recorded for the decision.	Functional	Medium	Business	AC-015: Status changes to 'Approved' or 'Rejected'; remarks are stored. Test: TC-Loan-02	Depends on BMS-F-014
BMS-F-016	The system shall notify the customer of the loan decision by email or SMS.	Functional	Low	Customer	AC-016: Notification sent within 1 minute of decision. Test: TC-Loan-03	Requires notification gateway

4.6 Statements, Reporting & Administration

This section covers customer statements as well as the reporting and administration features used by managers and administrators. It includes statement generation, staff account management, and reports for transactions, account openings, and loans.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test ref	Comments/Dependencies
BMS-F-017	The system shall generate a mini-statement showing the last 10 transactions and a full statement for a selected date range for any active account.	Functional	Medium	Customer	AC-017: Statement matches ledger entries for the selected range. Test: TC-Rep-01	Depends on transaction module
BMS-F-018	The system shall allow an Administrator to create, edit, deactivate, and assign roles to staff user accounts.	Functional	High	Admin	AC-018: Deactivated staff cannot log in. Test: TC-Admin-01	RBAC dependency
BMS-F-019	The system shall allow a Branch Manager or Administrator to generate daily or periodic reports covering transactions, account openings, and loan status.	Functional	Medium	Management	AC-019: Report totals reconcile with underlying transaction records. Test: TC-Rep-02	Exportable as PDF/CSV

5. Non-functional requirements (detailed)

The non-functional requirements describe the expected performance, reliability, security, accessibility, and data-retention levels of the system. Each requirement is measurable and linked to a test or verification method. These requirements are numbered from BMS-NF-001 onward.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
BMS-NF-001	The end-to-end response time for any account-level transaction (deposit, withdrawal, transfer) shall be ≤ 3 seconds for 90% of requests under normal load (up to 200 concurrent users).	Performance	High	90th percentile $\leq 3s$ in load test. Test: TC-Perf-01
BMS-NF-002	The system shall provide 99.5% availability during banking hours (9:00–18:00), excluding scheduled maintenance windows.	Reliability	High	Uptime monitoring reports $\geq 99.5\%$ monthly. Test: Ops reports
BMS-NF-003	All customer PII and financial data must be encrypted at rest and in transit; passwords and PINs must never be stored in plaintext.	Security/Compliance	High	Security audit / code review checklist pass. Test: TC-Sec-01
BMS-NF-004	The system shall retain audit logs of all financial transactions and login events, timestamped, for a minimum of 5 years.	Audit/Data Retention	High	Automated archival and retrieval verified. Test: TC-Ops-01
BMS-NF-005	The user interface shall conform to WCAG 2.1 AA accessibility guidelines and support both desktop and mobile viewports (responsive design).	Usability/Accessibility	Medium	Accessibility audit pass on key screens. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

- Confidentiality — Customer personal and financial information should only be available to users who are authorized to view it. Access to this information must be controlled by the user's role and permissions.
- Integrity — Account balances, transaction records, and loan decisions should only be changed through valid system operations. Any sensitive change should be recorded in the audit logs.
- Accountability — Sensitive activities such as logins, transactions, loan approvals, and configuration changes should be traceable to the user who performed them through the audit logs.
- Availability — Authorized users should be able to access the main banking functions within the defined uptime targets. The system should also remain stable when faced with reasonable levels of unexpected or malicious traffic.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test ref
BMS-SR-001	TLS 1.2 or higher shall be mandatory for all network connections between client, application server, and database.	Security	High	AC-SR-001: Connections on TLS < 1.2 are rejected. Test: TC-Sec-02
BMS-SR-002	Passwords shall be stored using a salted, one-way cryptographic hash (e.g., bcrypt/Argon2); plaintext passwords shall never be logged.	Security	High	AC-SR-002: DB inspection shows only hashed values. Test: TC-Sec-03
BMS-SR-003	The system shall enforce OTP-based two-factor authentication for login and for high-value transactions above a configurable threshold.	Security	High	AC-SR-003: Actions blocked without valid OTP. Test: TC-Sec-04
BMS-SR-004	The system shall log all authentication attempts, financial transactions, and administrative changes with user ID, timestamp, and outcome, and protect logs from tampering.	Security	High	AC-SR-004: Logs are append-only and reviewable by Admin. Test: TC-Sec-05
BMS-SR-005	The system shall automatically terminate an idle session after a configurable timeout (default 10 minutes) and require re-authentication.	Security	Medium	AC-SR-005: Session expires and protected pages redirect to login after timeout. Test: TC-Sec-06

6. Quality attributes & Acceptance tests

Exit criteria for acceptance

The project will be considered ready for acceptance when all High-priority functional requirements have been implemented and verified, there are no unresolved critical or high-severity NFR or security issues, and all linked test cases in the Requirements Traceability Matrix in Section 8 have passed.

Acceptance test suites

- Authentication & Access Control
- Account Opening & KYC
- Deposit & Withdrawal
- Funds Transfer
- Loan Management
- Reporting & Administration
- Performance
- Security
- Accessibility

7. System models and diagrams

7.1 UML Use-Case Diagrams

The document includes two use-case diagrams. The first shows the activities available to a Customer, while the second shows the activities performed by the Bank Teller, Branch Manager, and System Administrator.

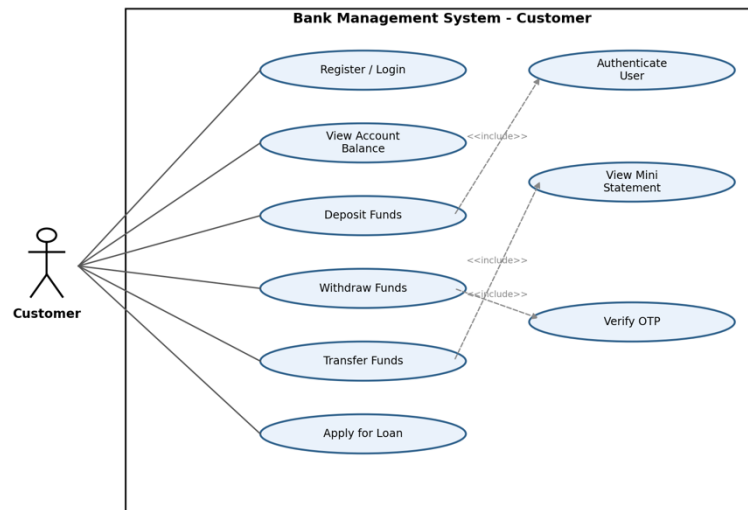


Figure 1: Use-case diagram - Customer interactions with the Bank Management System.

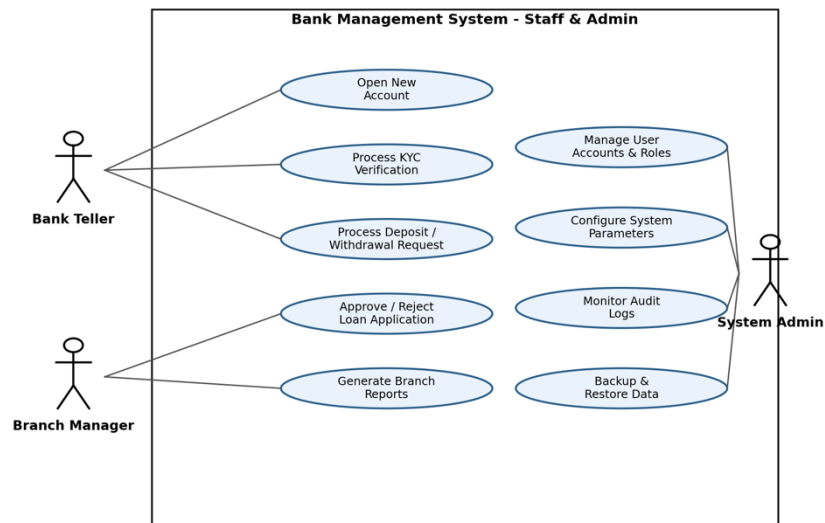


Figure 2: Use-case diagram - Bank Teller, Branch Manager, and System Administrator interactions.

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)
BMS-F-001	Customer registration	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N
BMS-F-002	Login + OTP	4.1 / DS-Auth-02	AuthModule	TC-Auth-02	N
BMS-F-003	Account lockout	4.1 / DS-Auth-03	AuthModule	TC-Auth-03	N
BMS-F-004	RBAC enforcement	4.1 / DS-Auth-04	AuthModule	TC-Auth-04	N
BMS-F-005	Open new account	4.2 / DS-KYC-01	AccountModule	TC-KYC-01	N
BMS-F-006	KYC approval routing	4.2 / DS-KYC-02	AccountModule	TC-KYC-02	N
BMS-F-008	Record deposit	4.3 / DS-Txn-01	TransactionModule	TC-Txn-01	N
BMS-F-009	Withdrawal limit check	4.3 / DS-Txn-02	TransactionModule	TC-Txn-02	N
BMS-F-011	Intra-bank transfer	4.4 / DS-Xfer-01	TransactionModule	TC-Xfer-01	N
BMS-F-012	OTP for high-value transfer	4.4 / DS-Xfer-02	TransactionModule	TC-Xfer-02	N
BMS-F-014	Loan application submission	4.5 / DS-Loan-01	LoanModule	TC-Loan-01	N
BMS-F-015	Loan approval/rejection	4.5 / DS-Loan-02	LoanModule	TC-Loan-02	N
BMS-F-017	Statement generation	4.6 / DS-Rep-01	ReportingModule	TC-Rep-01	N
BMS-F-018	Staff account management	4.6 / DS-Admin-01	AdminModule	TC-Admin-01	N
BMS-NF-001	Response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N
BMS-NF-003	Encryption at rest/transit	5 / DS-Sec-01	Security	TC-Sec-01	N
BMS-SR-001	TLS 1.2+ enforced	5.1.2 / DS-Sec-02	Security	TC-Sec-02	N
BMS-SR-003	OTP 2FA	5.1.2 / DS-Sec-04	Security	TC-Sec-04	N

Status legend: N = Not tested, P = Pass, A = As-designed/pending review.